AyarLabs

Ayar Labs Series B Discussion

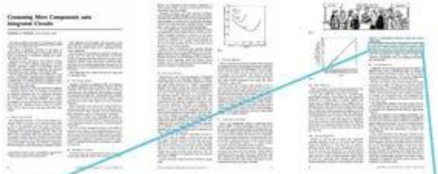
August 2020

Charlie Wuischpard, CEO

Lisa Cummins, CFO

The big industry computing challenge
"How do we get more out of Moore's?"

Gordon Moore had some ideas in the same paper in which he proposed Moore's Law:

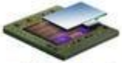


It may prove to be more economical to build large systems out of smaller functions, which are separately packaged and interconnected.

Split functionality onto multiple "chiplets" in a multi-chip package

Intel® AgileX™

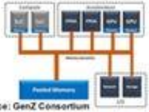
nVIDIA Tesla V100



Xilinx Virtex-7 HT

AMD Radeon R9 Fury X

Design disaggregated computing systems



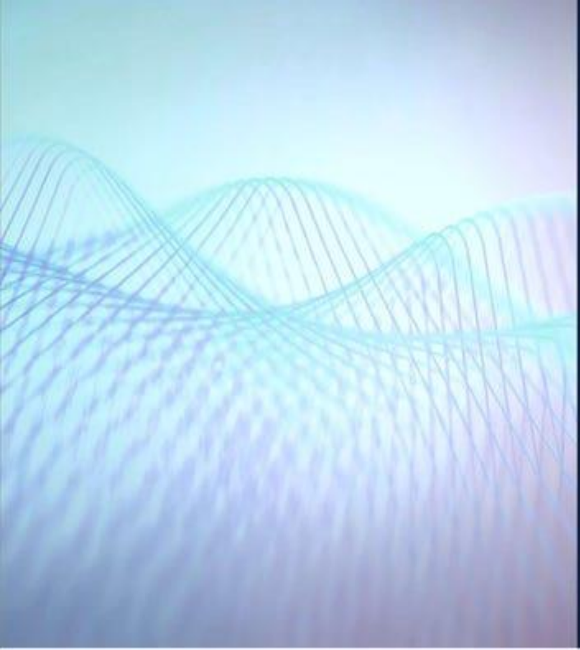
Source: GenZ Consortium

Key I/O requirements for disaggregated systems

- Fat pipes - *high bandwidths*
- Fast transmission - *low latency*
- Low power - *energy efficiency*

We Design and Sell Optical I/O

As a monolithic electronic/photonic chiplet-based solution for in-package I/O providing orders of magnitude improvement in bandwidth, bandwidth density and energy/distance



"Today, we know what technologies are necessary for the first and second generation of Exascale platforms in the 2022 to 2023 timeframe, but after that a crossover to optical I/O based solutions will be needed."

- Matt Leininger
Senior Principal HPC Strategist
Advanced Technology Office
Lawrence Livermore National Laboratory

"OpenAI must be on the cutting edge of AI capabilities and low latency, high bandwidth optical interconnect is a central piece of our compute strategy to achieve our mission of delivering artificial intelligence technology that benefits all of humanity."

- Chris Berner
Head of Compute
OpenAI

The Opportunity

Within 5-7 Years most data center short distance physical interconnects will be optical.

This represents a multi-billion-dollar market opportunity and Ayar Labs is the current leader for in-package optics ("the holy grail") as measured by technology and manufacturing readiness, and ecosystem traction

Meet The Team

Charlie Wuischpard
Chief Executive Officer
25+ years in semiconductor & compute. VP/GM Intel HPC. CEO Penguin Computing

Mark Wade
CTO & Co-Founder
Lead optical device inventor for world's first processor with optical I/O, MIT researcher, PhD

Lisa Cummins Dulchinos
Chief Financial Officer
25+ years growth experience
CFO Penguin Computing
CFO Adept Technology

Ken Chang
Senior VP of Engineering
20+ Years high speed links engineering. VP Wired Eng Xilinx. Stanford PhD. IEEE Fellow

Vladimir Stajanovic
Chief Architect, Co-Founder
UC Berkeley Professor of EECS
Co-Founder of NanoSemi
Ex-MIT Professor, Stanford PhD

Chen Sun
Chief Scientist, Co-Founder
VP of Silicon Engineering
Lead chip designer for world's first processor with optical I/O
UC Berkeley researcher, MIT PhD

Roy Meade
VP of Manufacturing
20+ years in semiconductor process engineering
Program Manager Micron

Hugo Saleh
VP of Marketing & Business Development
20+ years in semiconductor & compute
Sr. Director Intel Datacenter Sales

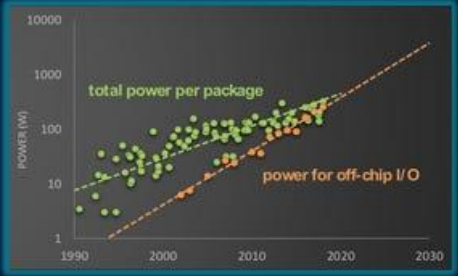
Alex Wright Gladstein
Chief Strategist & Co-Founder
Energy Entrepreneurship Lead MIT. Big data analytics at EnerNOC

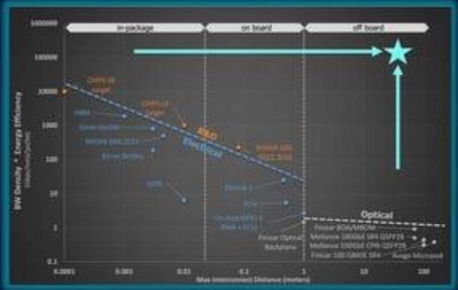
The Case for in-package optical I/O

- Package performance is pin limited
- Power for off-chip I/O increasing and unsustainable

- Heterogeneous datacenter workloads require new architectures mixing CPU, GPU, FPGA, Accelerator
- Resource pooling key to avoid stranded resources

- Large penalties for leaving the chip package and the board
- ~4 orders of magnitude difference in Figure of Merit from in-package to off board!





[G. Keeler, DARPA ERI 2019]

Multi-Chip Packages a Key Enabler of Optical I/O

Intel® AgileX™

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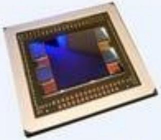
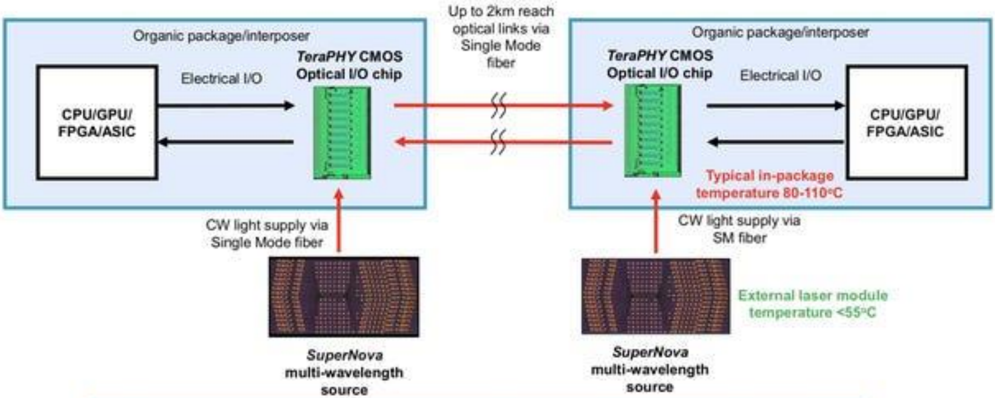


AMD Radeon R9 Fury X

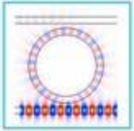
- ✓ Allows for mixed die function (GPU, CPU, memory, I/O, etc.)
- ✓ Established interconnects and packaging
- ✓ Optimize chiplet process for function and yield/cost (16nm, 10nm, DRAM, I/O, etc.)
- ✓ **For optics to use ecosystem, must be like electronics!**

Ayar Labs Optical I/O System Architecture

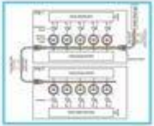


1000x the Bandwidth density at 1/10th the power and latency of copper
And at distances of up to 2km

Required for Optical I/O Chiplets

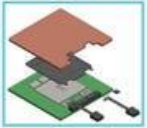


Dense optics: Micro-ring modulators



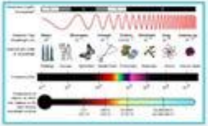
Monolithic integration with electronics

High volume scalable manufacturing



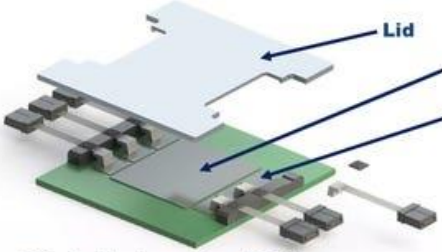
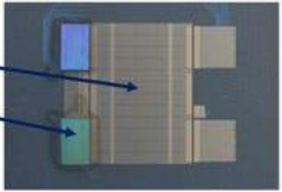
Advanced packaging and fiber attach techniques

Disaggregated multi-wavelength laser sources



Ecosystem and standardization: electrical I/O, wavelengths, data rates

Ayar Labs: Monolithic In-Package Optical I/O

Chiplet Integration Platform

- Bandwidth Density & Total Bandwidth Capacity (100X+)
- Power Consumption – Over Distance (<5pj/b)
- Reach (2KM)
- Error Free Transmission – enables new architectures

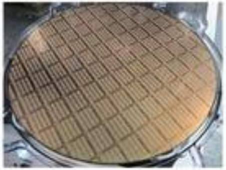
Industry First Demonstration – March 25 2020

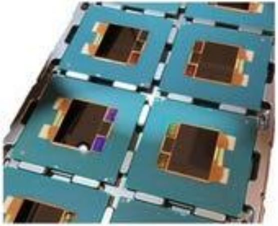
Defense Advanced Research Projects Agency   News And Events

PIPES Researchers Demonstrate Optical Interconnects to Improve Performance of Digital Microelectronics

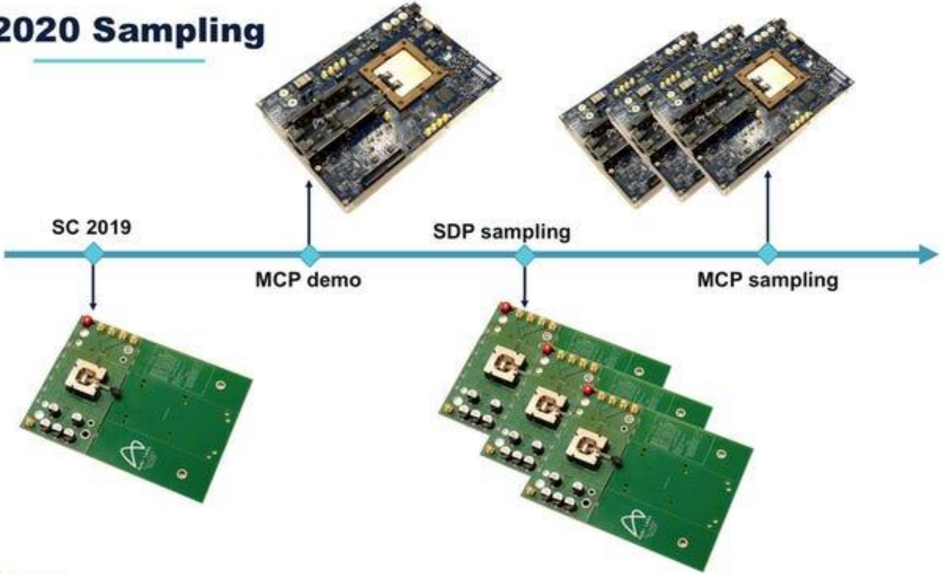
Researchers replace traditional electronic I/O with optical signaling interfaces to achieve major improvements in link reach and efficiency

OUTREACH@DARPA.MIL
3/25/2020





2020 Sampling



AyarLabs 10/15/2019 Confidential & Proprietary

Market Verticals and Application Areas

Aerospace & Government

Phased array radar
Digital beam forming

AI and HPC

Glueless interconnects
Memory semantic fabrics

Cloud and Telco

Disaggregated base station
Rack scale design

LIDAR

Next Gen Computing
Machine vision, Neuromorphic, Quantum

5-Year TAM > $10B in Data Center Alone

- **CPU/GPU/Accelerator/Networking Roadmaps will demand in-package optical I/O**
  - I/O requirements increasing
  - Power to move data unsustainable
  - Assumes 4-10 chiplets/server and 8+/switch
- **Enables New Architectures to accelerate Moores Law**
  - Resource Pooling & Peering
  - Disaggregated Memory
- **Assumes Price/GB for Ayar Photonic Chiplet < Copper costs**
- **Large Additional TAM Opportunities exist outside of the DataCenter**
  - Telecommunications (5G/6G)
  - Consumer Electronics (AR/VR)
  - Aerospace and Military

Appendix

Financial Model and Assumptions